

Heated Mattress: Test Guide from Unheated Raw Mattress to Closed-Loop PI

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How to use this guide

Run the phases in order and do not start one until the previous gate is met. The goal is to take the raw, unheated mattress at room temperature to a PI-controlled surface holding 37 °C within ± 0.3 °C, using `PI_tests_4_onward.ino` and `fopdt_fit.py`.

This guide characterizes a bench prototype; it does not qualify the device for patient use. Values marked *estimate* are engineering estimates, not measurements of your mattress. Tests 1 to 3 from the earlier plan (calibration, fault injection, thermostat bench check) are skipped here; Phases 2 and 8 cover the parts that matter on a live mattress.

1. **Phase 0, inspect (unplugged)**. Gate: no loose wire, exposed mains or damage.
2. **Phase 1, bench setup**. Gate: sensor reads sensibly and logging works.
3. **Phase 2, ambient baseline**. Gate: all sensors agree within 0.5 °C and are stable for 30 min.
4. **Phase 3, first power-on (Test 4)**. Gate: full power measured, no trips, nothing hot or smelly.
5. **Phase 4, open-loop steps (Tests 5 and 6)**. Gate: fit error below 0.1 °C and k within 30% between runs.
6. **Phase 5, enter gains and dry run**. Gate: no oscillation, warm-up overshoot below 1 °C.
7. **Phase 6, setpoint step (Test 7)**. Gate: overshoot below 0.5 °C, ripple within ± 0.3 °C.
8. **Phase 7, uniformity and disturbance (Tests 8 and 9)**. Gate: spots within ± 1 °C, recovery in the time you accept.
9. **Phase 8, live fault tests (Test 10)**. Gate: heater off within 5 s in every case.
10. **Phase 9, 6 h endurance (Test 11)**. Gate: no faults, temperature held within limits.

Safety rules and abort criteria

Every powered test is a live 120 V test, so these rules apply to every session.

Rules

- Two people present: one watches the mattress, one runs the laptop. No unattended runs until Phase 8 passes.
- At the start of each session, press the GFCI TEST button. It must trip; then press RESET.
- No bare mains conductors: SSR terminals and splices stay inside an insulated enclosure or are fully covered.
- Unplug from the wall before touching any wiring. Unplugging USB is not enough.
- Keep water, drinks and cleaning liquids away from the plug, SSR and splices. Test 9 uses a sealed, double-bagged container on top of the cover only.
- Never cut or loosen the braided ground. Keep neutral and ground continuous.
- Do not fold, stack or cover the heated section during tests, and do not pinch the wires.
- Keep an extinguisher rated for electrical fires (ABC or CO2) within reach.
- Have a qualified person check any mains rewiring before it is powered.

Abort at once (send o, then unplug from the wall) if any of these happen

- The surface sensor reads 40 °C or higher. The sketch latches off at 41 °C.
- The plate probe reads above 60 °C (estimate). Replace this with your thermostat rating minus 10 °C once you have chosen a thermostat.
- The GFCI trips.
- The SSR LED is on, or the watt meter shows power, while the log says duty 0. Treat it as a stuck-on SSR.
- Smell, smoke, crackling, discoloration, or a hot spot you did not expect.
- The log shows a fault code or nan readings.
- Anyone is unsure.

Equipment and materials

Gather everything below before Phase 1; the first three rows you already have.

Item	Used for	Approx. cost
Mattress with heater, inline GFCI plug, SSR in an insulated enclosure	Device under test	owned

Item	Used for	Approx. cost
Arduino Uno, USB cable, laptop with Arduino IDE and Python	Control, logging, curve fit	owned
<code>PI_tests_4_onward.ino</code> and <code>fopdt_fit.py</code> (<code>pip install numpy scipy</code>)	Firmware and analysis	free
Adafruit MAX31865 breakout, Pt1000 version	Reads the Pt1000	~\$18
Class A Pt1000 sensor, flat film or small probe	Control sensor	~\$5–15
Plug-in watt meter	Test 4 and idle-power checks	~\$20
Digital thermometer with K-type bead probe (two inputs if possible)	Plate and spot readings	~\$20
Multimeter	Continuity, resistance, isolation checks	owned or ~\$20
Kapton or aluminum tape, cable ties	Bonding probes, strain relief	~\$5
Room thermometer and stopwatch	Ambient and timing	owned
Sealed 1–2 L bottle or double-bagged water, thermos or kettle	Test 9 disturbance	~\$5
10 k Ω resistor	Pull-down on the SSR control pin (Phase 8)	under \$1
Printed log sheets (template at the end)	Hand-recorded plate temperatures	free
Optional: KSD301 thermostats, IR thermometer	Backup cutoff, surface scan	~\$1 each, ~\$15

Phase 0: Inspect everything unplugged

Goal: confirm the wiring, insulation and heater are sound before any power. Unplug the mattress from the wall and keep it unplugged for this phase.

1. Inspect the cord, GFCI body, cover and zipper for cuts, kinks, melted spots or exposed copper.
2. Check the wiring against your relay procedure: wall-side black to SSR terminal 1(–), ~~device-side black to terminal 2(–)~~. Yellow and the braided ground must be uncut and continuous.
3. Tug each screw terminal gently. No loose strands, no insulation under a screw, no bare conductor outside the enclosure.
4. Confirm the SSR is bolted flat to a metal plate or heat sink, following its datasheet derating curve.
5. Measure heater resistance with the multimeter between the device-side black wire (after the SSR) and the yellow wire. Expect about $16\ \Omega$ if the 7.38 A figure is real ($120\text{ V} \div 7.38\text{ A}$; estimate). An open circuit or a value far from this means stop and investigate.
6. Measure isolation: heater conductor to the braided ground should read open (OL, or above $1\text{ M}\Omega$). Any lower reading means a short to ground.
7. Measure the ground bond: the plug's ground pin to the braid should show continuity.
8. Check that the copper sheet is fully inserted, the mylar is flat, and no wire is pinched in the shell.
9. Install the Arduino library `Adafruit MAX31865 library`, upload `PI_tests_4_onward.ino` over USB, and open the Serial Monitor at 115200 baud with line ending Newline.
10. Type `?`. It should print settings with mode 0 (off) and no fault, and the SSR LED must stay off.

Gate: resistance within about $\pm 20\%$ of expectation, no isolation or ground fault, and the sketch runs with the heater idle. Write the measured resistance in the log; Phase 3 cross-checks it.

Phase 1: Set up the bench, probes and logging

Goal: fix every sensor position and the data path so each later test is repeatable. Do not move the probes between tests.

Room and mattress

- Use a draft-free room at 20–22 °C, away from vents, windows and sunlight. Record the room temperature.

- Lay the mattress flat and fully unfolded, head and foot sections in place, on a non-flammable table or floor.

Sensors (mark each position with tape so it never moves)

- **Pt1000 (control sensor):** tape it flat to the top of the cover at the center of the middle section, midway between two wire runs. Cover it with the folded sheet or drape you will use in every test, so it reads what a patient would contact.
- **K-type probe 1 (plate):** with the shell open, tape the bead to the heater plate near the center. Route the lead out through the zipper edge without pinching it, then close the shell.
- **K-type probe 2 (optional):** reserve it for the Test 8 spot readings on the cover.
- Keep the Pt1000 lead at least 10 cm from the mains wires, and cross them at 90° if it must cross.

Logging

- Close the Arduino Serial Monitor; only one program can hold the port. Log with `screen -L -Logfile run.csv <port> 115200` (macOS or Linux; find the port with `ls /dev/tty.*`), or PuTTY with Session > Logging set to All session output. You can still type commands in either.
- Name files `YYYYMMDD_test_description.csv`, for example `20261006_test5_step10.csv`. Start a new file for every test.
- Turn off laptop sleep and keep it plugged in. Lines starting with `#` are messages; the analysis script ignores them.

Gate: the log shows `temp_raw` near room temperature, `sensor_hex` 00 and `fault` 0 on every line, and the commands `o` and `?` respond.

Phase 2: Ambient baseline from the raw mattress

Goal: prove the mattress is at thermal equilibrium and the sensors agree, so every later temperature rise is caused by the heater. This is also a free single-point check of the Pt1000 against the K-type.

1. Leave the mattress unpowered in the test room for at least 2 h; overnight is better. Foam and plate must reach room temperature, or step tests start with a drift.
2. Place the probes (Phase 1), connect USB only, start a new log (`..._baseline.csv`), and keep the mode OFF (`o`).
3. Log for 30 min (about 900 lines at 2 s each).

- Record the Pt1000 average, plate K-type, room thermometer and the time.
- From the log, compute the `temp_filt` change over the last 10 min and the `temp_raw` peak-to-peak.

Check	Pass
Drift over last 10 min	below 0.1 °C
<code>temp_raw</code> peak-to-peak	below 0.2 °C
Pt1000, plate K-type, room thermometer	within 0.5 °C of each other
<code>sensor_hex</code> and <code>fault</code>	00 and 0 on every line

Most K-type meters are only accurate to about ± 1 °C (check yours), so this catches gross errors, not the ± 0.2 °C requirement. If the Pt1000 is off by X °C against a better reference, set `CAL_OFFSET = -X` in the sketch and re-upload. Do the ice-bath and 37 °C bath calibration before you claim ± 0.2 °C accuracy.

Gate: all four checks pass. Save this file; its start temperature is your T0 for every later cool-down comparison.

Phase 3: First power-on, power and idle check (Test 4)

Goal: measure true heater power and confirm the SSR is fully off when idle, before any heating test. This is a 30 s run.

- Plug the watt meter into the wall and the mattress into the meter. Press the GFCI TEST button (it must trip), then RESET.
- With the sketch in OFF mode, the meter must read about 0 W and the SSR LED must be off. More than a few watts means a leaking or stuck SSR: unplug and stop.
- Start a new log (`..._test4_power.csv`). Type `w`.
- The SSR is on for 0.6 s of every 2 s window. The meter will flicker; record its average over the last 15 s.
- After 30 s the sketch prints `AUTO DONE: power test finished` and the heater turns off. Confirm the meter returns to about 0 W.
- Compute full power: average watts $\div$ 0.30. Cross-check it against 120 V squared divided by the Phase 0 resistance.

7. Unplug from the wall, then touch-check the SSR enclosure and wire splices. They should be at most slightly warm.

Setting the duty cap. Start with `u` such that $\text{cap} \times \text{full power}$ is about 200 W (estimate; commercial underbody mattresses use far less). For an 886 W heater that is `u23`. The sketch's hard limit is 50%.

Gate: measured full power within $\pm 15\%$ of the resistance-based value, no GFCI trip, idle power about 0 W. Record full power and the chosen `u`.

Phase 4: Open-loop step tests (Tests 5 and 6)

Goal: heat at a fixed, known power and record how the surface responds. The fit gives dead time θ , time constant τ and gain k , which set the PI gains.

Before each run

- The mattress is within 0.3 °C of the Phase 2 baseline. After any earlier run, cool for at least 2 h with the heater off.
- Probes unmoved, door closed, same drape, nobody sitting or leaning on the mattress.
- New log file started, with 5 min logged at heater off for a clean baseline.

Test 5: 10% step

1. Type `s10` and note the clock time. Keep the log running.
2. Every 5 min write down the plate K-type reading (time and °C). Do not touch the mattress.
3. Watch the log. Expect a delay of minutes before the surface moves, then a slow climb (estimate).
4. The sketch stops itself and prints why: temperature settled (under 0.1 °C change over 10 min, after at least 20 min), the 90 min limit, or 39 °C. At 40 °C it latches a fault instead.
5. Keep logging 10 min after it stops, then let the mattress cool for at least 2 h.
6. Run `python fopdt_fit.py 20261006_test5_step10.csv` and record θ , τ , k , the RMS error and the percent of final rise reached.

Adjusting the step size

- Less than 0.5 °C rise at 90 min: repeat at `s20` or `s30`.
- More than about 8 °C rise within 20 min: repeat at `s5`, because the plate is heating fast.

Test 6: 20% step. Repeat the whole procedure with `s20` into a new file, then compare `k` with Test 5.

Check	Pass
Fit RMS error	below 0.1 °C
Log reached at least 85% of the fitted rise	yes; otherwise run longer
<code>k</code> from the 10% and 20% runs	within 30% of each other
Plate temperature at the end of each run	below 60 °C (estimate)

If `k` differs by more than 30%, the mattress is nonlinear. Use the run whose final surface temperature is closest to 37 °C for the gains. Record the plate temperature at the end of each run; it tells you how hot the plate must get to hold a given surface temperature.

Gate: both fits pass. Keep the best `step*.csv` files; they are your plant model.

Phase 5: Enter the gains and do a dry run

Goal: load the fitted `Kc` and `Ti`, then check closed-loop behavior in a short attended run before the full setpoint test.

1. Run the fit with the default smooth tuning: `python fopdt_fit.py file.csv`. Use `python fopdt_fit.py file.csv 1` for faster, riskier tuning only after the smooth version works.
2. Type the printed values into the terminal, for example `k0.0558` then `i599`. Set the cap from Phase 3, for example `u23`. Type `?` to confirm.
3. Sanity check: `Kc` should fall between about 0.005 and 0.5 duty per °C, and `Ti` between 60 and 3600 s (estimates). Outside that range, re-check the fit.
4. Start a new log (`..._dryrun.csv`) and type `t34`. The sketch does not accept targets below 30 °C.
5. Stay with the mattress for the first 30 min. Watch `duty_pct`, `temp_filt` and `I_pct`. The duty will sit at the cap during warm-up; that is normal.
6. Abort with `o` if `temp_filt` exceeds 35 °C (1 °C above target), or if the duty swings between 0 and the cap repeatedly.

Gate: warm-up overshoot below 1 °C and no sustained oscillation. The settings reset when the Arduino resets, so once you are happy, copy `Kc` , `Ti` and `U_MAX` into the sketch defaults and re-upload.

Phase 6: Closed-loop setpoint step (Test 7)

Goal: show the controller holds 34 °C, steps to 37 °C without overshoot, then holds 37 °C steadily. One command runs it.

1. Confirm gains and cap with `?` . Start a new log (`..._test7_34_37.csv`).
2. Type `a` . The controller warms to 34 °C and waits until `temp_filt` stays within ± 0.2 °C for 10 min. If it cannot settle within 2 h it stops and says why.
3. It then logs `# MARK setpoint step 34 -> 37` and runs for 60 min. The setpoint ramps at 0.5 °C/min, so the step itself takes about 6 min.
4. At 60 min it prints overshoot, minutes to within ± 0.5 °C, and the temperature statistics for the last 20 min. The heater keeps holding 37 °C until you send `o` .

Result	Pass
Overshoot above 37 °C	below 0.5 °C
Ripple, $(\max - \min) \div 2$ over the last 20 min	within ± 0.3 °C
Minutes to within ± 0.5 °C	your choice; I suggest under 30
Faults	none

Tuning rules. Change one parameter by about 25% per iteration. Then repeat from `t34` , wait for steady state, and send `t37` .

What the log shows	Likely cause	Change
Endless swings lasting tens of minutes	Kc too high or Ti too small	Lower Kc, or raise Ti
Slow, never quite reaches target	Kc too low or Ti too large	Raise Kc, or lower Ti
Overshoot above 0.5 °C	Integral too aggressive	Raise Ti, or lower Kc

What the log shows	Likely cause	Change
Stuck 0.3 to 0.5 °C low for a long time	Integral too slow	Lower Ti
Duty pinned at the cap, temperature below target	Cap too low or sensor lag	Raise <code>u</code> in 5% steps (max 50%)

Allow at most four or five tuning iterations. If it still fails, the cause is more likely sensor position or dead time than the gains; revisit the Phase 1 placement before tuning further.

Gate: both numeric criteria pass in one clean run.

Phase 7: Uniformity and disturbance (Tests 8 and 9)

Goal: confirm the whole surface is near the sensor reading, and that the controller recovers when something cold lies on the mattress.

Test 8: Uniformity

1. Hold `t37` until `temp_filt` is within ± 0.2 °C for 20 min.
2. Record the plate K-type reading; it is the plate temperature needed to hold 37 °C at the surface.
3. Tape the K-type bead flat to the cover, under the drape, at five spots in turn. Leave each for 3 min before reading.
4. Compare each reading with the Pt1000.

Spot	K-type reading (°C)	Difference from Pt1000 (°C)
Center		
Near the head edge		
Near the foot edge		
Directly over a wire run		
Midway between two runs		

Pass: each spot within ± 1 °C of the average and none above 38 °C. The ± 1 °C figure comes from a summary of the heating-mattress standard (IEC 80601-2-35); confirm it in the current edition.

Test 9: Disturbance

1. At steady 37 °C for 20 min, start a new log (`..._test9_waterbag.csv`).
2. Warm 1 to 2 L of water to about 35 °C in a sealed bottle or double-bagged container, and wipe it dry.
3. Type `b`, then place it on top of the drape over the Pt1000 position. Keep it well away from the plug, SSR and splices.
4. Watch the dip and recovery for 30 min. Note the lowest temperature and the time to return to within ± 0.5 °C.
5. Type `b` again when you remove it. Log 30 more min and note any overshoot.

Pass: it returns to within ± 0.5 °C in a time you accept (I suggest 15 min or less), and overshoot after removal is below 0.5 °C. If duty sits at the cap while the temperature stalls, raise `u` by 5% and repeat.

Phase 8: Live fault tests (Test 10)

Goal: prove that a failed sensor or a dead controller stops the heater quickly. Run with two people and the mattress holding 37 °C.

1. **Add the pull-down first.** With the mattress unplugged, fit a 10 k Ω resistor between Arduino pin D7 and GND. It keeps the SSR input low if the Arduino resets or loses power.
2. **Sensor unplug.** With `t37` running, unplug the Pt1000 from the breakout. Within about 4 s the log shows `# FAULT 1`, the SSR LED goes dark, and the watt meter falls to about 0 W. Reconnect, then send `r`. It must refuse while the sensor is unplugged and accept once it is back.
3. **Sensor short.** Briefly short the two Pt1000 terminals at the breakout. Expect fault 1 or 2 and the same heater shutoff.
4. **Controller loss.** Unplug the USB cable while heating. The SSR LED must go dark and the meter must fall to about 0 W within 2 s.
5. **Idle power.** After each test, confirm the meter reads about 0 W while the sketch is in OFF mode.

Do not force the SSR on to simulate a stuck relay. The sketch cannot catch a shorted SSR; only a series thermostat or contactor can. Install and bench-test one before the long run in Phase 9.

Test	Pass
Sensor unplugged	Heater off within 5 s, fault 1 logged
Sensor shorted	Heater off within 5 s, fault 1 or 2
USB removed	Heater off within 2 s
Idle after each test	Meter about 0 W

Gate: all four pass.

Phase 9: Six-hour endurance (Test 11)

Goal: show the controller holds 37 °C for longer than a typical surgery, with no faults or drift. Start only after Phases 6 to 8 pass, because this is the longest powered run.

1. Confirm gains and cap are saved in the sketch defaults. Start a new log (`..._test11_endurance.csv`). Disable laptop sleep and plug the laptop in.
2. Type `t37` . When `temp_filt` is within ± 0.3 °C of 37, type `z` to zero the statistics.
3. Stay for the first hour. After that, check every 30 min and keep someone in the building. Do not leave the mattress in an empty room.
4. Every 30 min the sketch prints cumulative min, max, mean and sd. Copy them to the log sheet with the plate K-type and room temperatures.
5. Optionally repeat the Test 9 water-bag disturbance once, at about 3 h.
6. At 6 h send `o` , unplug from the wall, and inspect the cover, wire insulation, SSR enclosure and plug for discoloration, softness or melted spots.

Check	Pass
Cumulative max <code>temp_filt</code> after <code>z</code>	at most 38.0 °C
Cumulative min <code>temp_filt</code> after <code>z</code>	at least 36.0 °C
Cumulative sd	below 0.15 °C (my suggestion)
Fault codes	none
Post-run inspection	no damage or discoloration

The 2000-cycle durability requirement is a separate test and is outside this guide.

Pass/fail summary

Nine checks gate the move from a raw mattress to a controlled one.

Phase	Test	Command	Pass criterion
2	Baseline	<code>o</code>	Drift below 0.1 °C per 10 min; sensors within 0.5 °C
3	4, power	<code>w</code>	Full power within ±15% of V^2/R ; idle about 0 W
4	5, 10% step	<code>s10</code>	Fit RMS error below 0.1 °C
4	6, 20% step	<code>s20</code>	k within 30% of Test 5
6	7, setpoint step	<code>a</code>	Overshoot below 0.5 °C; ripple within ±0.3 °C
7	8, uniformity	manual	Spots within ±1 °C; none above 38 °C
7	9, disturbance	<code>b</code>	Back within ±0.5 °C in the time you accept
8	10, faults	manual	Heater off within 5 s in every case
9	11, endurance	<code>t37</code> , <code>z</code>	36.0 to 38.0 °C, no faults

Troubleshooting

Symptom	Likely cause	Action
<code>sensor_hex</code> not 00, or <code># FAULT 1</code> at start	Pt1000 wiring, 2-wire jumper, loose terminal	Check RTD+ and RTD-. For 2-wire, tie F+ to RTD+ and F- to RTD- per Adafruit's guide
Reads about -240 °C or very high	Sensor shorted or open	Check the leads; the sketch faults (code 2)
Wrong by tens of degrees	Wrong board or reference resistor	Use the Pt1000 board (4300 Ω reference); the sketch sets RNOMINAL 1000 and RREF 4300
<code>nan</code> in the log	No good reading yet	Wait for 3 good reads; check wiring

Symptom	Likely cause	Action
w, s or t refused	Fault latched or sensor not ready	Read the # line; send r after fixing the cause
Meter shows power in mode 0	Stuck or leaky SSR	Unplug from the wall; replace the SSR
GFCI trips	Leakage to ground, damp or damaged insulation	Unplug; repeat the Phase 0 isolation check
Fault 3 during a run	Loose sensor lead or electrical noise	Re-tape the sensor; reroute the lead away from mains
Fault 5 during a run	Sensor detached, or heater not heating	Re-tape the sensor; confirm the heater warms the plate
Step test ends at 90 min with tiny rise	Duty too low	Repeat at a higher percentage
Swings in temperature	Gains too high	Use the Phase 6 tuning table
Arduino keeps resetting	Watchdog on an old bootloader	Set <code>USE_WATCHDOG</code> to 0 and re-upload

Suggested schedule

Allow about four working sessions; all durations are estimates and depend mostly on how long the mattress takes to cool.

Session	Contents	Time
1	Phases 0 to 2, then Test 4	3 h work, plus a 2 h soak before the baseline
2	Test 5, 2 h cool-down, Test 6	5 to 6 h
3	Phase 5, Test 7, then Tests 8 and 9	5 to 7 h
4	Phase 8 fault tests, then Test 11 endurance	8 h

Test log template

Copy this table into a new tab or onto paper and fill one row per run.

Test	Date	Log file	Kc / Ti / cap	Result	Notes
4, power					
5, 10% step					
6, 20% step					
7, setpoint step					
8, uniformity					
9, disturbance					
10, faults					
11, endurance					